Dhyey Desai

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CAREER OBJECTIVE

Third-year Computer Science student with hands-on computer vision experience in anatomy segmentation. IBM-certified in Machine Learning, interested in AI/ML, deep learning, and CV, and eager to apply AI skills to solve real-world problems.

EDUCATION

B.Tech, Computer Science & Engineering 2023 - 2027
Sarvajanik College of Engineering & Technology

Senior Secondary (XII), CBSE 2023
Ryan International School, Surat

Percentage: 92.00%

TRAININGS / CERTIFICATIONS

Infosys Virtual Internship 7.0 AI Track
Apr 2026

infosys, Virtual

Completed the Artificial Intelligence learning track through the Infosys Springboard Virtual Internship Program 7.0, gaining hands-on exposure to AI, Data Science, Deep Learning, Natural Language Processing (NLP), Computer Vision, Generative AI, and Prompt Engineering and various other concepts through industry-oriented coursework and practical learning modules.

Data Science & Machine Learning Course
Dec 2024 - Dec 2025

GeeksforGeeks, Virtual

Completed the Complete Machine Learning & Data Science Program from GeeksforGeeks, where I learned the fundamentals of Python, data analysis, machine learning, and AI. Gained hands-on experience in data cleaning, preprocessing, exploratory data analysis, and building predictive models using different machine learning algorithms. Worked with tools and technologies such as SQL, Excel, Power BI, and Python libraries for data science. Also explored advanced topics including deep learning and NLP through practical problems

PORTFOLIO

GitHub link ↗

Portfolio link ↗

PROJECTS

IoT Based Car Theft Detection System ↗
Apr 2026

Developed an IoT-based smart vehicle security and theft detection prototype using GPS and real-time monitoring systems to enhance parked vehicle safety. Integrated GPS sensor tracking with predefined threshold-based detection logic to identify suspicious vehicle movement, unauthorized displacement, and potential theft attempts. Designed an automated alert mechanism that instantly sends location updates and security notifications to the user's mobile application for real-time monitoring and rapid response. Implemented sensor-driven decision logic to minimize false alerts while improving detection accuracy under different environmental and parking conditions. Worked on embedded system integration, live location tracking, and communication workflows to simulate a scalable intelligent anti-theft solution for smart transportation systems.

Automated Anatomy Segmentation from Laparoscopic Surgical Images ↗
Sep 2025 - Present

This project is being pursued by Dhyey Desai under the guidance of faculty Prof. (Dr.) Rachana Oza.
This project develops an advanced surgical anatomy segmentation system using transformer-based deep learning models, progressing from UNET & Swin-UNet to a custom Swin-UNETR, DeepLabV3. Built with PyTorch, OpenCV, and GPU-accelerated mixed-precision training, it accurately segments multiple anatomical structures from real surgical images. The pipeline includes rigorous multi-phase training, quantitative and qualitative evaluation, and anatomy-level reasoning from predicted masks. This work supports future AI-assisted surgery, medical training, and clinical decision support.

SKILLS

- Python
- NumPy
- Pandas
- Matplotlib
- Machine Learning
- Neural Networks
- Data Analysis
- Prompt Engineering
- Artificial intelligence
- Data Science
- Feature Engineering
- PyTorch
- TensorFlow
- Keras
- Computer Vision
- Effective Communication
- GitHub
- Research and Analytics
- scikit-learn
- Model Evaluation
- OpenCV
- Deep Learning

EXTRA CURRICULAR ACTIVITIES

- Participated in Google's Gen AI Exchange Hackathon organized in January 2026
- Participated in Google Solution Challenge in 2025
- Participated in Google Cloud's Agentic AI Day Hackathon
- Participated in the international NASA Space Apps Challenge 2024 and 2025
- Volunteered in organizing the event of "Origami Art" of the annual college event of Exuberance in the year 2025

ADDITIONAL DETAILS

- Secured 1st position in the OEP Poster Presentation and won cash prize
- Secured 1569 rank in the Amazon ML Challenge 2025 out of approximately 23000 teams